

Why Have All the Doctors Gone? Insights Into Early Clinical Departure Among Physicians in the United States: A National Survey

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Abstract

INTRODUCTION: This national survey of clinically inactive physicians was conducted to identify the factors driving early exit from the clinical physician workforce in the United States, aiming to evaluate characteristics and motivations for leaving early.

METHODS: A sample of clinically inactive physicians drawn from American Medical Association Physician Professional Data™ completed a survey between May and June 2024, with questions assessing demographics, education, clinical training, and reasons for leaving clinical practice. In addition to standard descriptive statistics, gender differences were also explored.

RESULTS: Among the 971 respondents included in the analysis, the majority (63.9%) identified as women, the mean age was 45.8 years, and 11.0% had never practiced after graduate medical education. The physicians who left practice reported “hassle factor” (44.7%) and “too stressful” (44.5%) as prime motivators for their departure. Physicians who were women were more likely than men to have exited due to needing to care for family members (7.9% vs 0.6%, $P < .001$) or children (21.3% vs 4.2%, $P < .001$). The mean age of physicians who left clinical practice was 48.1 years, 9 years younger than observed in a similar cohort in 2008.

DISCUSSION: This study suggests that physicians who have left practice early have had shorter clinical careers than in the past. Interventions to reduce “hassle factor” and workplace stress may address the motivations for leaving practice.

CONCLUSION: Understanding early attrition from clinical practice may improve interventions in sustaining the physician workforce. Specifically, further study is needed for women physicians who were fully trained but never entered clinical practice, as these groups will likely contribute an outsized effect on the magnitude of the workforce shortage.

Introduction

The clinical physician workforce supply involves a complex relationship between the production of newly trained physicians, the pool of actively practicing physicians, and the eventual exit of those physicians from clinical practice.¹ In recent decades, insufficient numbers of medical graduates² and higher numbers of exiting physicians^{3,4} have tipped the balance of this relationship, resulting in an impending deficiency in the US physician workforce.

The Association of American Medical Colleges has projected a shortage of between 13,500 and 86,000 physicians in the United States by the year 2036.⁵ To mitigate the projected shortage, emphasis has been placed on retaining the current workforce by reducing burnout,⁶ improving organizational factors,⁷ and increasing the number of residency training positions.^{2,8} In addition to these important factors, the loss of physicians from the clinical workforce, particularly if the rate of loss exceeds the rate of entry by new physicians, must be considered.

Many factors contribute to a physician's decision to move jobs or leave clinical practice altogether. Drivers of burnout and other factors associated with intent to leave practice have recently been studied in detail.⁹⁻¹⁵ Although this evidence presents valuable insights into the sentiments of practicing physicians, intent to leave does not necessarily translate into actually leaving clinical practice. Stakeholders and policymakers also must understand the motivations of physicians who have already left the practice of clinical medicine earlier than the traditional retirement age, including those who finished training but never practiced.

Various authors have studied the drivers of early exit from clinical practice for specific physician groups (eg, specialty-specific, geographic regions).¹⁶⁻¹⁸ However, the authors know of only 1 US national survey, completed in 2008, which studied early exit across all physicians.¹⁹ Since that time, the clinical practice landscape has changed dramatically (eg, wide adoption of electronic health records (EHRs), passage of the Health Information Technology for Economic and Clinical Health (HITECH) and Affordable Care Acts, and a global pandemic), perceptions of US health care have changed,^{20,21} attention

to physician mental health and well-being has increased,²² a new generation of physicians has been trained, and more nonclinical educational and career opportunities are available. Therefore, drivers of early physician attrition from the workforce, defined as exiting clinical practice before the traditional retirement age, may have changed substantially, and more timely evidence is needed to tailor interventions.

This national survey of clinically inactive physicians was conducted to identify the factors driving early exit from the clinical physician workforce in the United States. The study aimed to evaluate information about the demographic, educational, and nonclinical career characteristics of physicians who are no longer practicing clinical medicine as their primary vocation as well as motivations for their exit.

Methods

PARTICIPANTS

A population of 64,801 physicians across all specialties, who graduated residency between 2000 and 2022 (inclusively), was selected from American Medical Association Physician Professional Data™ (AMA PPD) to represent a population of physicians who had clinical career lengths of less than 25 years. To reflect physicians most likely not currently practicing clinical medicine, the population was also chosen to only include all unlicensed physicians and licensed physicians listed with “unclassified” or “not in direct patient care” as the type of practice.

An initial email invitation to participate in an online survey was distributed May 17, 2024, followed by 4 reminder emails. A \$20 e-gift card incentive was offered for participation. Of the initial distribution, 3512 emails were returned undeliverable, leaving 61,289 contactable physicians. Survey responses received by June 14, 2024, were recorded and comprised 7358 physicians (12.0% response rate). After 6387 were screened out of eligibility (see criteria described in the following sections), 971 responses remained for analysis (1.6% of the original population).

Out of the eligible respondents, the reason for leaving practice was elicited from the 521 physicians no longer practicing clinical medicine, excluding those who had never practiced or were currently practicing.

Participation in the study was voluntary and responses were anonymous. Participants provided informed consent at the start of the survey. The University of Illinois Chicago institutional review board reviewed the study and determined it to be exempt from human participant research requirements (STUDY 2024-0368). Investigators followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines in the presentation of this study.

STUDY MEASURES

Screening questions were asked at the start of the survey to eliminate physicians who 1) at the time practiced patient care more than 20 hours per week; 2) graduated from graduate medical education before 2000 or after 2022; or 3) did not complete a residency program. For participants who were not screened out, the remainder of the survey included basic demographic questions and questions about education, clinical training, specialty, and career choices.

Questions to elicit reasons for leaving clinical practice and details about nonclinical career transitions were also included. Participants were asked, “What are the reasons you are no longer practicing clinical medicine?” and given a choice of multiple options for selection, which included those derived from the Physician Workforce Reentry questionnaire used to survey national “inactive” physicians in 2008,¹⁹ Shanafelt et al,³ and others developed for this study.

Respondents were also given the option to provide a free-text response to this question. Two of the authors [SC and MT] undertook qualitative thematic analysis of these responses, including cyclical coding.²³ Independently, each of these authors first reviewed all the free-text responses, then developed overall themes, and finally coded each response based upon theme. The authors then came to consensus on overall themes, jointly reviewed and deconflicted discordantly coded responses, and arrived at a consensus theme for each response. Participants were also asked to indicate what, if anything, would make them consider reentering clinical practice.

Please see the supplementary appendix for survey details.

STATISTICAL ANALYSIS

Following established practice, standardized mean differences (SMDs) were used to assess balance between the analytic sample and the target population using physician gender and age. Absolute SMDs

below 0.10 were interpreted as negligible imbalance, while values exceeding 0.20 were considered indicative of a meaningful lack of representativeness.^{24,25}

Standard descriptive statistics were used to demonstrate the composition of the study sample. Means and standard deviations were established for continuous measures, and frequency and proportional values were used for demonstrating categorical values. Simple Wald z tests using logistic regression, with no adjusting covariates, were performed to ascertain potential gender differences of respondents not currently practicing medicine.

To determine differences in clinical career lengths between physicians who are men and physicians who are women, a median survival analysis technique was employed to estimate the proportion of physicians still in clinical practice at any given career duration. Kaplan-Meier curves visually represent the probability of remaining in practice at various time points for both men and women physicians. A log-rank test was performed to evaluate whether there was a statistically significant difference in clinical career lengths between the 2 groups over the study period. Statistical significance was set at $P < .05$ for all analyses.

Missing data were minimal across demographic and educational variables (generally $< 3\%$ per item). Because the proportion and pattern of missingness were small and appeared random (ie, no systematic differences by gender, age, or practice status), listwise deletion was applied for descriptive and inferential analyses. No imputation procedures were used, as the degree of missingness was unlikely to meaningfully affect point estimates or associations.

Statistical analysis was completed using StataNow 19 SE statistical software, release 19 (StataCorp LLC).

Results

The initial population of 64,801 sourced from AMA PPD data comprised physicians with a mean age of 41.9 years, and 51.1% were women. The 7358 survey responders had a mean age of 40.5 years and 54.6% were women. The SMD for age was -0.17 and the SMD for gender (woman) was 0.07 , indicating the survey respondents were representative of the initial study population. The majority (63.8%) of the 971 eligible respondents identified as women, 76.3% were married, 60.8% were white, and the mean age was 45.8 years.

More than 1 out of 7 respondents graduated from a medical school outside the United States (14.5%), and a plurality (27.9%) completed their residency in an internal medicine or related specialty. More than half (51.5%) had completed a fellowship.

Three practice status categories emerged among the respondents, with 107 (11.0%) having never practiced after graduate medical education, 521 (53.7%) having practiced in the past, and 343 (35.3%) currently practicing part time (less than 20 hours per week).

Demographic and medical education characteristics of all eligible respondents, listed by practice status, are reported in Tables 1 and 2.

In the cohort of physicians no longer practicing clinical medicine, “hassle factor” (44.7%) and “too stressful” (44.5%) were the most frequently reported reasons for leaving clinical medicine, followed by “increasingly unrealistic patient demands” (41.1%) and “lack of professional satisfaction” (38.4%). The categorical responses are shown in Figure 1.

From the optional free-text answers, 252 coded responses were identified (multiple codes per respondent were allowed). The top 4 themes that emerged were overwork ($n = 44$), pursuing a nonclinical role ($n = 37$), frustration with health system structures (including organizational administration) ($n = 35$), and profession-related personal reasons (burnout, moral injury, lack of autonomy) ($n = 24$).

GENDER DIFFERENCES

Women respondents were more likely to indicate they left clinical practice to care for other family members at 7.9% vs 0.6% for men (odds ratio [OR], 0.13; 95% CI, 0.04–0.44), to care for young children at 21.7% vs 4.2% for men (OR, 0.29; CI, 0.18–0.48), due to health concerns at 13.8% vs 3.8% (OR, 0.47; CI, 0.28–0.81), and because it was too stressful at 31.7% vs 12.9% (OR, 0.65; CI, 0.45–0.94). Inference was based on Wald z statistics, and estimated odds ratios with associated confidence intervals are displayed in Figure 2.

Physicians who were women were also found to have fewer years in clinical practice (median 9 years) than

physicians who were men (median 12 years; $P = .006$). Figure 3.

Discussion

SHORTER CLINICAL CAREERS

This study suggests that physicians in this clinically inactive population were younger than a similar sample of clinically inactive physicians previously studied. The mean age of physicians in this sample was 9 years younger than the mean age of 57.1 years observed in a similar cohort in 2008.¹⁹ Additionally, a small but nontrivial proportion of this study's sample population consisted of physicians who never practiced clinical medicine after completing residency. To date, the authors do not know of any reports addressing the size or characteristics of this population of physicians. Identifying drivers of attrition for this completely trained but never practicing physician group, including but not limited to burnout,²⁶ will be important to shape interventions aimed at keeping residents engaged and helping them transition successfully into clinical practice.

This study highlights the importance of considering the age of physicians when they leave practice, given that a key parameter in physician workforce supply projections is the average retirement age. Any changes in this parameter can greatly change workforce estimates. For example, in the latest Association of American Medical Colleges physician workforce projections, a decrease in the average retirement age by 2 years would result in a decrease of the projected physician supply in 2036 by approximately 40,000.⁵

National rates of attrition from the clinical workforce are difficult to measure but are estimated to have risen from 1.7% in 2010 to 3.1% in 2018.⁴ The authors' study does not contribute directly to increasing the precision of these attrition estimates, as they did not address the relative size of this clinically inactive cohort with respect to the entire physician population. However, it is difficult to imagine that this proportion is getting smaller. Given that, and difficulties with measurement notwithstanding, increasingly shorter clinical careers, including for those who never

Personal demographics and medical education	Never practiced n = 107 (11.0%)		Not practicing n = 521 (53.6%)		Currently practicing n = 343 (35.3%)		Total sample N = 971	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Age, y	43.6	8.3	48.1	8.8	42.9	7.6	45.8	8.7
	n	Percent	n	Percent	n	Percent	n	Percent
Gender								
Missing	0	0.0	0	0.0	3	0.9	3	0.3
Man	44	41.1	175	33.6	110	32.1	329	33.9
Nonbinary	0	0.0	0	0.0	2	0.6	2	0.2
Prefer not to say	1	0.9	10	1.9	6	1.8	17	1.8
Woman	62	57.9	336	64.5	222	64.7	620	63.9
Race								
Missing	2	1.9	3	0.6	11	3.2	16	1.7
American Indian or Alaska Native	1	0.9	1	0.1	0	0.0	2	0.2
Asian	32	29.9	138	26.5	83	24.2	253	26.1
Black or African American	3	2.8	20	3.8	24	7.0	47	4.8
Other, please specify	10	9.4	29	5.6	20	5.8	59	6.1
Pacific Islander or Native Hawaiian	1	0.9	2	0.4	1	0.3	4	0.4
White	58	54.2	328	63.0	204	59.5	590	60.8
Relationship								
Missing	0	0.0	0	0.0	11	3.2	11	1.1
Married	60	56.1	403	77.4	278	81.1	741	76.3
Partnered	10	9.4	23	4.4	20	5.8	53	5.5
Single	36	33.6	94	18.0	32	9.3	162	16.7
Widowed/Widower	1	0.9	1	0.2	2	0.6	4	0.4
Year graduated medical school								
Missing	4	3.7	8	1.5	15	4.4	27	2.8
1990–1995	3	2.8	28	5.4	5	1.5	36	3.7
1996–2000	14	13.1	148	28.4	51	14.9	213	21.9
2001–2005	14	13.1	124	23.8	58	16.9	196	20.2
2006–2010	17	15.9	102	19.6	62	18.1	181	18.6
2011–2015	31	29.0	82	15.7	93	27.1	206	21.2
2016–2020	24	22.4	29	5.6	59	17.2	112	11.5
2021+	0	0.0	0	0.0	0	0.0	0	0.0
International medical graduate								
Missing	2	1.9	4	0.8	12	3.5	18	1.9
No	87	81.3	430	82.5	295	86.0	812	83.6
Yes	18	16.8	87	16.7	36	10.5	141	14.5
Residency specialty								
Missing	0	0.0	3	0.6	12	3.5	15	1.5
Anesthesiology	1	0.9	14	2.7	6	1.7	21	2.2
Dermatology	0	0.0	3	0.6	4	1.2	7	0.7
Emergency medicine	2	1.9	24	4.6	30	8.8	56	5.8

Table 1: Demographics and medical education (*Continued*)

Table 1: *Continued*

	n	Percent	n	Percent	n	Percent	n	Percent
Family medicine	5	4.7	63	12.1	36	10.5	104	10.7
Internal medicine or related specialty	23	21.5	144	27.6	104	30.3	271	27.9
Internship only	5	4.7	1	0.2	0	0.0	6	0.6
Neurology	3	2.8	20	3.8	11	3.2	34	3.5
Obstetrics and gynecology	5	4.7	25	4.8	11	3.2	41	4.2
Ophthalmology	0	0.0	7	1.3	5	1.5	12	1.2
Pathology	20	18.7	26	5.0	9	2.6	55	5.7
Pediatrics or related specialty	15	14.0	97	18.6	59	17.2	171	17.6
Physical medicine and rehabilitation	3	2.8	8	1.5	3	0.9	14	1.4
Preventive medicine, occupational medicine, or environmental medicine	9	8.4	14	2.7	4	1.2	27	2.8
Psychiatry	5	4.7	27	5.2	33	9.6	65	6.7
Radiation oncology	3	2.8	2	0.4	5	1.5	10	1.0
Radiology	4	3.7	8	1.5	5	1.5	17	1.8
Surgical specialty	4	3.7	35	6.7	6	1.8	45	4.6
Fellowship								
Missing	0	0.0	0	0.0	11	3.2	11	1.1
No	54	50.5	281	53.9	125	36.4	460	47.4
Yes	53	49.5	240	46.1	207	60.4	500	51.5
Year completed residency								
Missing	1	0.9	0	0.0	0	0.0	1	0.1
1990–1995	0	0.0	0	0.0	0	0.0	0	0.0
1996–2000	3	2.8	36	6.9	9	2.6	48	4.9
2001–2005	18	16.8	169	32.4	63	18.4	250	25.8
2006–2010	17	15.9	131	25.1	45	13.1	193	19.9
2011–2015	20	18.7	84	16.1	77	22.5	181	18.6
2016–2020	34	31.8	84	16.1	120	35.0	238	24.5
2021+	14	13.1	17	3.3	29	8.5	60	6.2

SD = standard deviation.

practice, would contribute to continued rise in attrition rates and a greater shortfall in the physician workforce.

MOTIVATIONS FOR LEAVING CLINICAL PRACTICE, THE RELATIONSHIP TO BURNOUT, AND POTENTIAL INTERVENTIONS

The results from the authors' study may signal a potential change in drivers of early attrition in the last 15 years. In this survey, the top 4 reasons reported for no longer practicing clinical medicine were different from those reported in the 2008 survey, in which the reasons were personal health issues (37.8%), hassle factor (32.5%),

rising malpractice premiums (24.4%), and lack of professional satisfaction (23.6%).

The top reasons given in the authors' study align with the components of burnout, which can be defined as the prolonged response to chronic workplace stressors and consist of emotional exhaustion, depersonalization, and lack of personal accomplishment.²⁷ It has become a major concern for physicians, with almost half of physicians in the United States experiencing burnout during their careers.⁶ Administrative burdens (ie, hassle factor) and unrealistic patient demands can contribute to overwork, stress, and decreased professional satisfaction.

Degree and career characteristics	Never practiced n = 107 (11.0%)		Not practicing n = 521 (53.7%)		Currently practicing n = 343 (35.3%)		Total sample N = 971	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Practice length	-	-	9.3	6.6	-	-	-	-
	n	Percent	n	Percent	n	Percent	n	Percent
Maintained medical license								
Missing	-	-	5	1.0	-	-	-	-
No	-	-	137	26.1	-	-	-	-
Yes	-	-	384	73.0	-	-	-	-
Work in nonclinical field (> 10 h) for compensation								
Missing	0	0	4	0.8	15	4.4	19	2.0
No	45	42.1	270	51.3	198	58.6	513	52.8
Yes	62	57.9	252	47.9	125	37.0	439	45.2
Other degrees								
Missing	3	2.8	19	3.6	24	7.1	46	4.7
No	49	45.8	284	54.0	133	39.4	466	48.0
Yes	55	51.4	223	42.4	181	53.6	459	47.3

Table 2: Other degree and career characteristics (N = 971)^a

^a Practice length and medical license information was not collected from the respondents who reported they are currently practicing or had never practiced, and therefore the corresponding statistics are shown in the table with an en dash.

SD = standard deviation.

Although literature on interventions to reduce burnout has grown,²⁸ the search continues for actionable solutions to the drivers of burnout.

For example, with respect to overwork, physicians are more likely to experience burnout when they do not take adequate time away from work.²⁹ Not taking time to rest and recharge because of barriers like struggling to find coverage or facing excessive inbox work upon return is associated with higher odds of leaving practice or reducing hours.¹³ Normalizing the expectation that physicians take time away from work and implementing policies and procedures that eliminate barriers to taking the time away involves a cultural shift for which organizations can strive.

In addition, organizations can build scheduling models that allow for flexible scheduling without penalties. Interventions to improve physician influence over their work environment (eg, flexible or customizable work schedules) can positively affect physician well-being,³⁰ and enabling physicians to have more control over their schedules can make a big difference in physicians' work-life balance.¹²

Unrealistic patient demands may include the desire to discuss multiple health problems in one visit

meant for an acute problem, demands for prescriptions without a consultation, the expectation that the physician is available to text message or answer phone calls at any time,³¹ large volumes of patient messages to their physician's inbox,³² or expectations that the physician can diagnose and begin a treatment immediately for every problem.³¹ Ensuring physicians are equipped with the right job resources for patient care,³³ have team support (eg, triaging patient requests),³⁴ and have adequate time for managing messages^{35,36} can help mitigate stress associated with managing patient demands.

Interestingly, EHR frustration was not one of the top reasons for exiting clinical practice reported by participants in this study, although other studies report that frustration with EHRs is a major factor contributing to physician burnout,³⁷⁻³⁹ and EHR frustration is just one of many reasons for exiting clinical practice.¹⁷ This may suggest that although interventions tailored to mitigate EHR frustration could help alleviate burnout, concentrating on the other drivers may be more impactful.

Finally, for physicians, passion and sense of purpose are integral to remaining in their job and engaged with their role as a doctor. For many, medicine is a calling, and burnout can erode that sense of

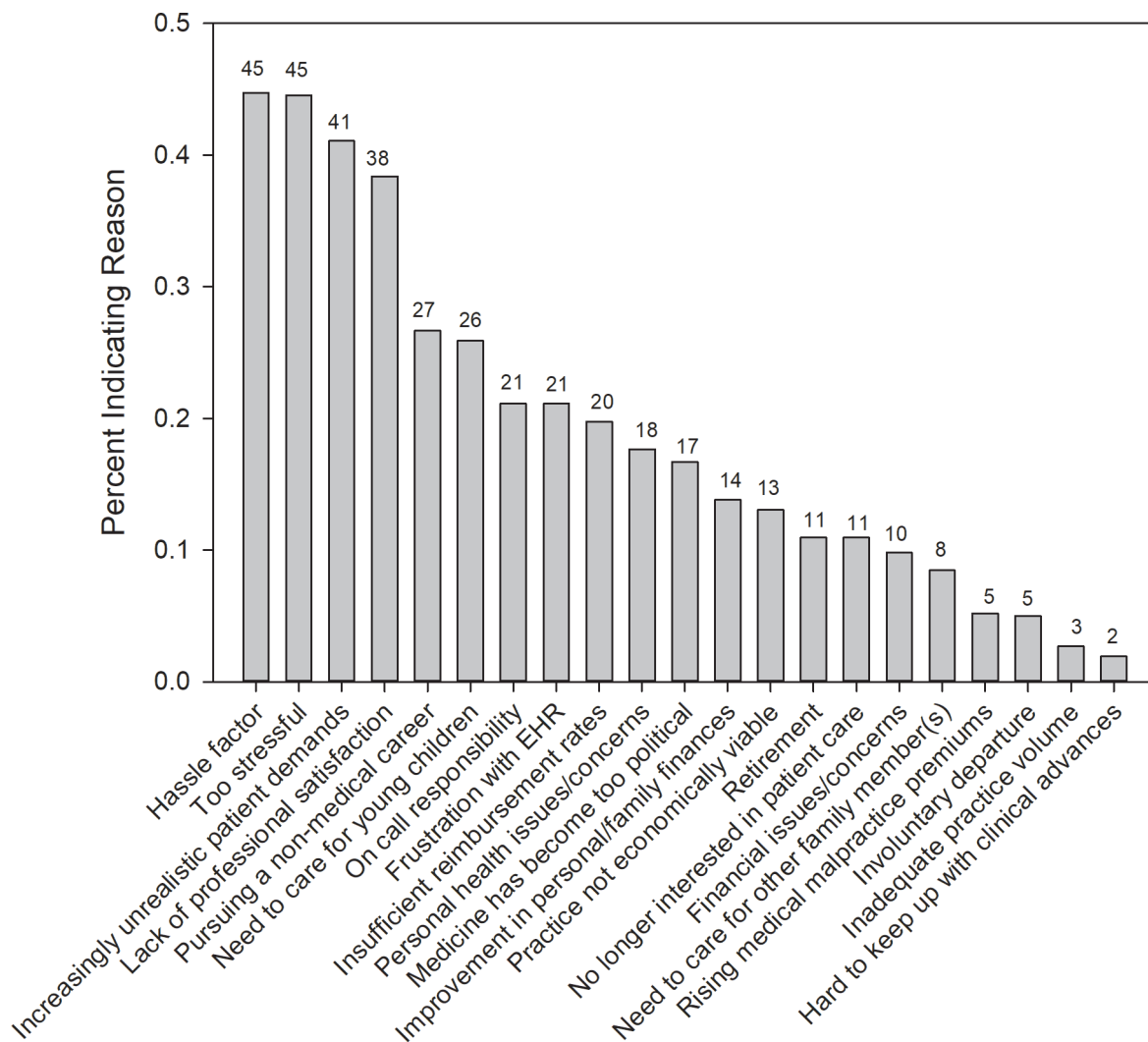


Figure 1: Reasons for being clinically inactive (N = 521). Notes: A total of 521 respondents were not currently practicing. The free-text “Other” option (n = 252 responses) is not represented in this figure (see results for further details). Finally, decimal points were omitted in the graph to simplify visual data. EHR = electronic health record.

calling,⁴⁰ possibly to the point that physicians leave medicine altogether. Organizations that tap into this sense of purpose and meaningfully empower physicians with control over aspects of their patient care can improve the environment for physicians.

In summary, the top drivers of early departure from clinical practice found in this study align with the components of burnout, although the authors are not necessarily suggesting a direct causative effect. Interventions to ease these drivers may also decrease burnout, and vice versa, potentially mitigating workforce attrition.

THE GENDER GAP

The authors found key differences between men and women physicians in the eligible respondent sample. Although physicians who were women comprised a bit over half the original AMA PPD eligible population and initial survey sample, the proportion of physicians who were women in the eligible respondent sample was nearly two-thirds. In addition, women in this study left clinical practice earlier than men, a finding mirrored in other studies.^{4,41} Furthermore, women in this study were more likely than men to exit the clinical workforce to care for

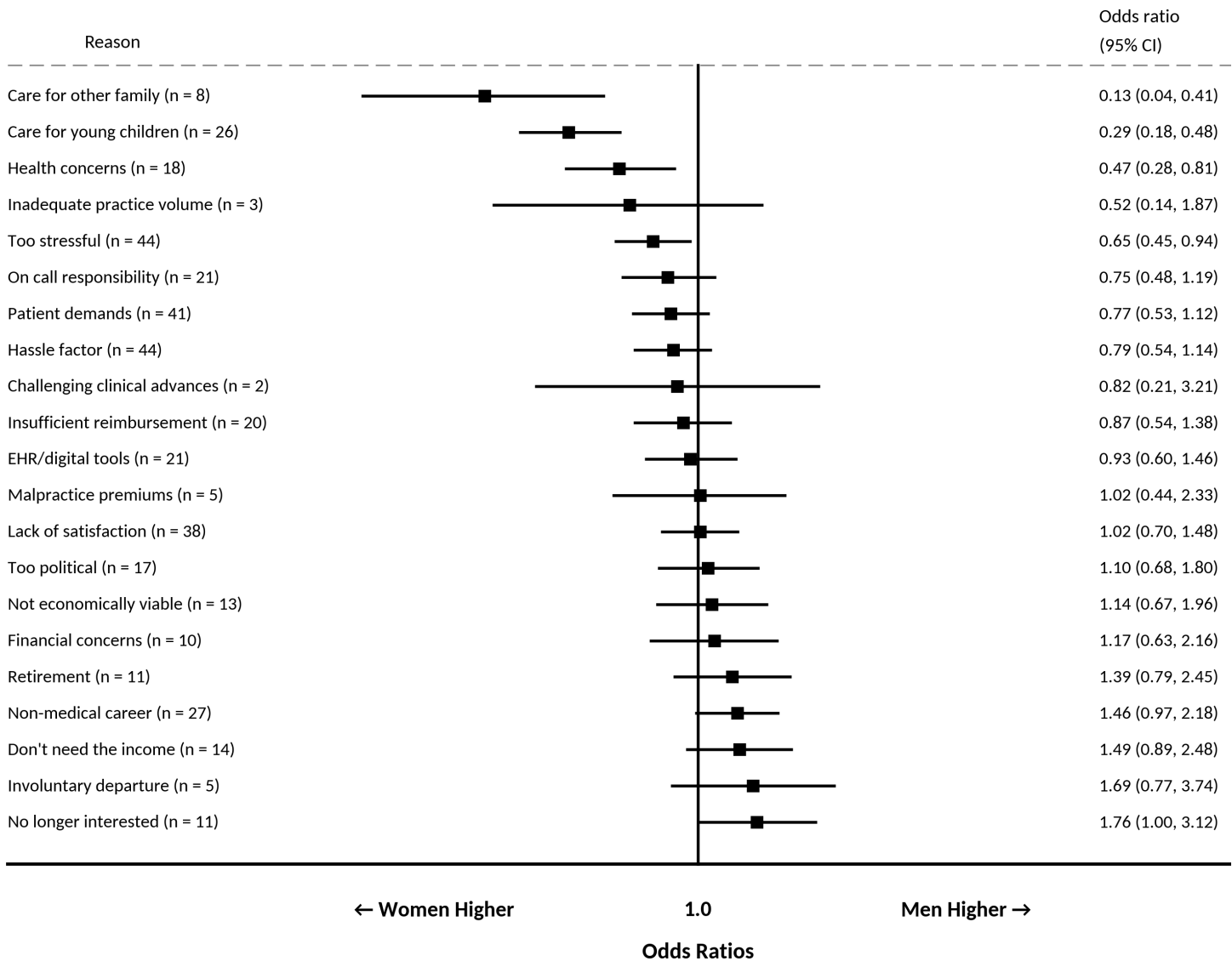


Figure 2: Forest plot for reasons for exit by gender. Each row represents a reason for exiting clinical practice, with the square indicating the estimated odds ratio based upon gender and the horizontal line representing the 95% CI. A vertical “no difference” line is shown along the unity hash (1.0) of the x axis. If the horizontal CI line for a particular reason crosses the vertical “no difference” line, there is no statistically significant difference between the genders. Notes: There were a total of 336 women and 185 men. Nonbinary and “prefer not to say” respondents were excluded from this analysis due to the small numbers of respondents in these groups. CI = confidence interval; EHR = electronic health record.

young children and other family members, for health concerns, and due to stress.

Considered together, the higher proportion of physicians who were women and their earlier departure from medicine may signal that the departure among physicians who were women may have a greater compound effect on the overall workforce than the departure of physicians who were men.

Furthermore, the proportion of clinically inactive physicians who are women will likely follow an upward trend as there has been a continuously increasing proportion of women entering US medical schools in the past decade; as of 2023–2024, women comprised 55.4% of matriculants.⁴² Similarly, the proportion of active residents who were women recently approached half (48.2%).⁴³ Addressing the drivers of early clinical exit that

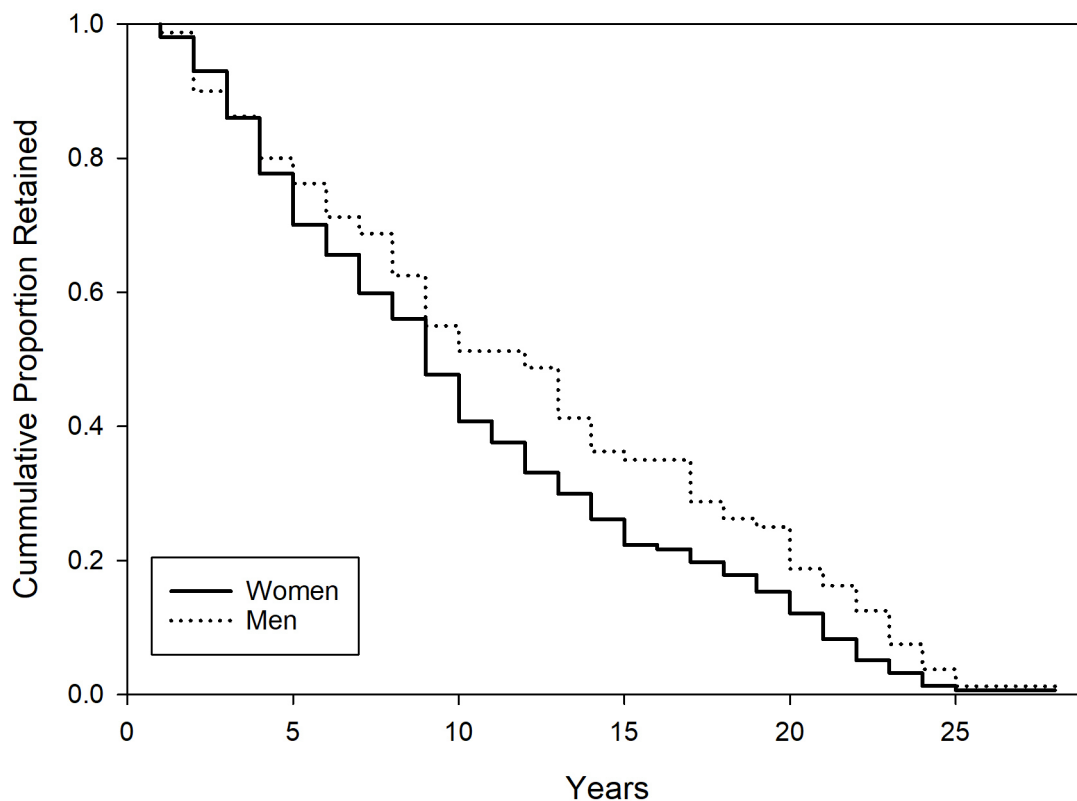


Figure 3: Kaplan-Meier plot for proportion of physician population remaining in clinical practice by gender. The Kaplan-Meier curves represent the estimated cumulative probability of remaining in clinical practice over the course of 30 years for physicians who are women (solid line) and physicians who are men (dotted line) in this study population that are not currently practicing ($N = 521$). The x axis represents the time (in years) from the start of a clinical career post-training. The steeper decline for physicians who are women reveals a rate of earlier clinical career exit as compared with physicians who are men. The difference in clinical career length for physicians who are women (median of 9 years) vs physicians who are men (median of 12 years) was statistically significant (log-rank, $P = .006$).

affect women will be increasingly important as their representation in the physician workforce continues to grow.⁴⁴

Women, in general, are more likely than men to remain out of the workforce to provide care outside of paid work and, for nearly 50% of women, caring for children full time is the primary reason they remain unemployed.⁴⁵ Physicians who are women and physicians with young children report feeling less supported by their organizations than physicians who are men and physicians without young children,⁴⁶ and other research shows that childcare stress is more likely for physicians who are women and is associated with both intent to leave practice and intent to reduce clinical hours.⁴⁷ This gender disparity may be improved by organizations that provide support for access to high-quality, affordable childcare, which may include onsite services

or partnering with reputable childcare centers to provide discounts on tuition for their employees. Family leave policies that provide adequate paid time away for parents with newborns or new children and support for lactating mothers, including high-quality private facilities and sufficient time away, can also improve employment experiences for many physicians who are women.

Physicians who were women in this study were also significantly more likely to report health concerns as a reason for clinical departure. Efforts can be made to ensure benefit plans include coverage for mammograms and other cancer screenings, reproductive and menopause-related health care, and dietary and fitness support.

Physicians who were women in this study were more likely to report leaving clinical medicine because it

was too stressful. Studies show that physicians who are women have fewer leadership roles, earn less money, and experience mistreatment and discrimination in their profession more often than physicians who are men,⁴⁸⁻⁵⁰ factors that likely disproportionately contribute to a stressful work environment. Evaluating and correcting any of these gender disparities is a first step and should be a top priority for all organizations.

Given that the physician workforce comprises an increasing proportion of women, and that it is likely that women are leaving clinical practice earlier and comprise a higher proportion of clinically inactive physicians, tailoring interventions which differentially impact women to a greater degree may lead to an amplified effect in decreasing total early attrition rates.

LIMITATIONS

This study is subject to limitations. The study population is a cohort of clinically inactive physicians who are likely to be disengaged with medicine, and the response rate was low, which may lead to response bias. Nonresponse testing was also not performed. The study population is not representative of US physicians, as it provides analysis from the retrospective perspective of physicians who have already left clinical practice. Therefore, the findings cannot be generalized as such to all physicians. The subjective nature of the survey questions and the self-reported survey responses introduce potential for bias.

Conclusion

To the authors' knowledge, this is the only report in well over a decade to detail national survey data from physicians, regardless of specialty, who are not currently practicing medicine as their primary vocation. Understanding the attributes of early attrition from the clinical workforce can help stakeholders and policymakers tailor interventions to support a more sustainable physician workforce.

The current motivational drivers of early exit from clinical careers may be different than those of the past. For example, in this study, unrealistic patient demands may now play a larger role than increasing malpractice insurance premiums or EHR frustrations. Allocation of resources for intervention will need to adapt to these changing needs.

Although difficult to measure on a national scale, decreasing lengths of clinical careers will negatively impact the physician workforce. The small but substantial number of physicians who complete residency but do not practice clinical medicine will contribute an outsized effect on the magnitude of the workforce shortage. Further study of this group is necessary to staunch this immediate leakage of these highly trained professionals.

There is a growing proportion of physicians who are women in the workforce. Addressing key differences in attrition drivers, including a disproportionate number of women exiting due to needing to care for children, other family members, or personal health issues, may lead to an amplified beneficial effect on the clinical workforce.

Data-Sharing Statement

The underlying data are currently unavailable.

Supplementary Materials

Supplemental material is available at: <https://www.thepermanentejournal.org/doi/suppl/10.7812/TPP/25.219#supplementary-materials>.

REFERENCES

1. Rittenhouse DR, Mertz E, Keane D, Grumbach K. No exit: An evaluation of measures of physician attrition. *Health Serv Res.* 2004;39(5):1571-1588. DOI: <https://doi.org/10.1111/j.1475-6773.2004.00304.x>
2. Lubell J. Congress revives bill to add 14,000 GME slots over seven years. *AMA News Wire*; July 16, 2025. Accessed March 31, 2026. <https://www.ama-assn.org/education/gme-funding/congress-revives-bill-add-14000-gme-slots-over-seven-years>
3. Shanafelt TD, Dyrbye LN, West CP, et al. Career plans of US physicians after the first 2 years of the COVID-19 pandemic. *Mayo Clin Proc.* 2023;98(11):1629-1640. DOI: <https://doi.org/10.1016/j.mayocp.2023.07.006>
4. Bond AM, Casalino LP, Tai-Seale M, et al. Physician turnover in the United States. *Ann Intern Med.* 2023;176(7):896-903. DOI: <https://doi.org/10.7326/M22-2504>
5. GlobalData Plc. The complexities of physician supply and demand: Projections from 2021 to 2036. Association of American Medical Colleges; March 2024.
6. Shanafelt TD, West CP, Sinsky C, et al. Changes in burnout and satisfaction with work-life integration in physicians and the general US working population between 2011 and 2023. *Mayo Clin Proc.* 2025;100(7):1142-1158. DOI: <https://doi.org/10.1016/j.mayocp.2024.11.031>
7. Olson K, Marchalik D, Farley H, et al. Organizational strategies to reduce physician burnout and improve professional fulfillment. *Curr Probl Pediatr Adolesc Health Care.* 2019;49(12):100664. DOI: <https://doi.org/10.1016/j.cpped.2019.100664>
8. Resident Physician Shortage Reduction Act of 2025, HR 3890, 119th Congress (2025-2026).

9. LeClaire M, Poplaur S, Linzer M, Brown R, Sinsky C. Compromised integrity, burnout, and intent to leave the job in critical care nurses and physicians. *Crit Care Explor.* 2022;4(2):e0629. DOI: <https://doi.org/10.1097/CCE.0000000000000629>
10. Mallick S, Douglas PS, Shroff GR, et al. Work environment, burnout, and intent to leave current job among cardiologists and cardiology health care workers: Results from the national Coping With COVID survey. *J Am Heart Assoc.* 2024;13(18):e034527. DOI: <https://doi.org/10.1161/JAHA.123.034527>
11. Rotenstein LS, Brown R, Sinsky C, Linzer M. The association of work overload with burnout and intent to leave the job across the healthcare workforce during COVID-19. *J Gen Intern Med.* 2023;38(8):1920–1927. DOI: <https://doi.org/10.1007/s11606-023-08153-z>
12. Sinsky CA, Brown RL, Rotenstein L, Carlasare LE, Shah P, Shanafelt TD. Association of work control with burnout and career intentions among U.S. physicians: A multi-institution study. *Ann Intern Med.* 2025;178(1):20–28. DOI: <https://doi.org/10.7326/ANNALS-24-00884>
13. Sinsky CA, Shah P, Carlasare LE, Shanafelt TD. Association between vacation characteristics and career intentions of US physicians—A cross-sectional analysis. *Mayo Clinic Proceedings.* 2025;100(5):814–827. DOI: <https://doi.org/10.1016/j.mayocp.2024.09.020>
14. Collins RT, Schadler A, Huang H, Day SB, Bauer JA. Impact of burnout and professional fulfillment on intent to leave among pediatric physicians: The findings of a quality improvement initiative. *BMC Health Serv Res.* 2024;24(1):434. DOI: <https://doi.org/10.1186/s12913-024-10842-2>
15. Ligibel JA, Goularte N, Berliner JL, et al. Well-being parameters and intention to leave current institution among academic physicians. *JAMA Netw Open.* 2023;6(12):e2347894. DOI: <https://doi.org/10.1001/jamanetworkopen.2023.47894>
16. Stajduhar T. On the verge of a physician turnover epidemic: Physician retention survey results — February 2021. Jackson Physician Search. 2021. Accessed April 2, 2026. https://www.jacksonphysiciansearch.com/wp-content/uploads/2021/02/Jackson-Physician-Search_2021-White-Paper_Physician-Retention-Survey-Results.pdf
17. O'Connell R, Hosain F, Colucci L, Nath B, Melnick ER. Why do physicians depart their practice? A qualitative study of attrition in a multispecialty ambulatory practice network. *J Am Board Fam Med.* 2023;36(6):1050–1057. DOI: <https://doi.org/10.3122/jabfm.2023.230052R2>
18. Oh RC, Mohr DC, Schult TM. VA physicians intent to leave and correlations to drivers of burnout: A cross-sectional study. *BMC Health Serv Res.* 2025;25(1):125.
19. Jewett EA, Brotherton SE, Ruch-Ross H. A national survey of “inactive” physicians in the United States of America: Enticements to reentry. *Hum Resour Health.* 2011;9(1):7. DOI: <https://doi.org/10.1186/1478-4491-9-7>
20. Brenan M. View of U.S. healthcare quality declines to 24-year low. Gallup. December 6, 2024. Accessed April 2, 2026. <https://news.gallup.com/poll/654044/view-healthcare-quality-declines-year-low.aspx>
21. Saad L. Historically Low Faith in U.S. Institutions Continues. Gallup. July 6, 2023. Accessed April 2, 2026. <https://news.gallup.com/poll/508169/historically-low-faith-institutions-continues.aspx>
22. Shanafelt TD. Physician well-being 2.0: Where are we and where are we going? *Mayo Clinic Proceedings.* 2021;96(10):2682–2693. DOI: <https://doi.org/10.1016/j.mayocp.2021.06.005>
23. Braun V, Clarke V. *Thematic Analysis: A Practical Guide.* Sage; 2021.
24. Austin PC. Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity—Score matched samples. *Statistics in Medicine.* 2009;28(25):3083–3107. DOI: <https://doi.org/10.1002/sim.3697>
25. Stuart EA. Matching methods for causal inference: A review and a look forward. *Stat Sci.* 2010;25(1):1–21. DOI: <https://doi.org/10.1214/09-STS313>
26. Shanafelt TD, Dyrbye LN, Sinsky C. Changes in burnout and satisfaction with work-life integration among U.S residents and fellows and the general U.S. working population between 2012 and 2023. *Acad Med.* 2012. DOI: <https://doi.org/10.1097/ACM.00000000000006294>
27. Maslach C, Leiter MP. Understanding the burnout experience: Recent research and its implications for psychiatry. *World Psychiatry.* 2016;15(2):103–111. DOI: <https://doi.org/10.1002/wps.20311>
28. Panagioti M, Panagopoulou E, Bower P, et al. Controlled interventions to reduce burnout in physicians: A systematic review and meta-analysis. *JAMA Intern Med.* 2017;177(2):195–205. DOI: <https://doi.org/10.1001/jamainternmed.2016.7674>
29. Sinsky CA, Trockel MT, Dyrbye LN, et al. Vacation days taken, work during vacation, and burnout among US physicians. *JAMA Netw Open.* 2024;7(1):e2351635. DOI: <https://doi.org/10.1001/jamanetworkopen.2023.51635>
30. Dunn PM, Arnetz BB, Christensen JF, Homer L. Meeting the imperative to improve physician well-being: Assessment of an innovative program. *J Gen Intern Med.* 2007;22(11):1544–1552. DOI: <https://doi.org/10.1007/s11606-007-0363-5>
31. Lateef F. Patient expectations and the paradigm shift of care in emergency medicine. *J Emerg Trauma Shock.* 2011;4(2):163–167. DOI: <https://doi.org/10.4103/0974-2700.82199>
32. Nath B, Williams B, Jeffery MM, et al. Trends in electronic health record inbox messaging during the COVID-19 pandemic in an ambulatory practice network in New England. *JAMA Netw Open.* 2021;4(10):e2131490. DOI: <https://doi.org/10.1001/jamanetworkopen.2021.31490>
33. Lee RS, Son Hing LS, Gnanakumaran V, et al. INSPIRED but tired: How medical faculty's job demands and resources lead to engagement, work-life conflict, and burnout. *Front Psychol.* 2021;12:609639. DOI: <https://doi.org/10.3389/fpsyg.2021.609639>
34. Halstater B, Kuntz M-C. Using a nurse triage model to address patient messages. *Fam Pract Manag.* 2023;30(4):7–11.
35. Mandal S, Wiesenfeld BM, Mann DM, Szerencsy AC, Iturrate E, Nov O. Quantifying the impact of telemedicine and patient medical advice request messages on physicians' work-outside-work. *NPJ Digit Med.* 2024;7(1):35. DOI: <https://doi.org/10.1038/s41746-024-01001-2>
36. Akbar F, Mark G, Prausnitz S, et al. Physician stress during electronic health record inbox work: In situ measurement with wearable sensors. *JMIR Med Inform.* 2021;9(4):e24014. DOI: <https://doi.org/10.2196/24014>
37. Budd J. Burnout related to electronic health record use in primary care. *J Prim Care Community Health.* 2023;14:21501319231166921. DOI: <https://doi.org/10.1177/21501319231166921>
38. Tai-Seale M, Baxter S, Millen M, et al. Association of physician burnout with perceived EHR work stress and potentially actionable factors. *J Am Med Inform Assoc.*

- 2023;30(10):1665–1672. DOI: <https://doi.org/10.1093/jamia/ocad136>
39. Tajirian T, Stergiopoulos V, Strudwick G, et al. The influence of electronic health record use on physician burnout: Cross-sectional survey. *J Med Internet Res*. 2020;22(7):e19274. DOI: <https://doi.org/10.2196/19274>
40. Jager AJ, Tutty MA, Kao AC. Association between physician burnout and identification with medicine as a calling. *Mayo Clin Proc*. 2017;92(3):415–422. DOI: <https://doi.org/10.1016/j.mayocp.2016.11.012>
41. Rotenstein LS, He Z, Dziura J, et al. Sex differences in physician attrition from clinical practice across specialties: A nationwide, longitudinal analysis. *J Gen Intern Med*. Published online April 2, 2026. DOI: <https://doi.org/10.1007/s11606-026-10362-1>
42. Association of American Medical Colleges. Table A-7.2: Applicants, first-time applicants, acceptees, and matriculants to U.S. MD-granting medical schools by gender, 2014–2015 through 2023–2024. 2014. Accessed October 27, 2023. <https://www.aamc.org/media/9576/download>
43. Brotherton SE, Etzel SI. Graduate medical education, 2022–2023. *JAMA*. 2023;330(10):988–1011. DOI: <https://doi.org/10.1001/jama.2023.15520>
44. Boyle P, Dill M, Kelly R, Nouri Z. Women are changing the face of medicine in America. *AAMCNews*. May 28, 2024. Accessed April 5, 2026. <https://www.aamc.org/news/women-are-changing-face-medicine-america>
45. Wielk E. Women in the Workforce Need Family-Focused Policy. *Bipartisan Policy Center*. October 10, 2023. Accessed April 5, 2026. <https://bipartisanpolicy.org/article/women-in-the-workforce-need-family-focused-policy/>
46. Carlasare LE, Wang H, West CP, et al. Associations between organizational support, burnout, and professional fulfillment among US physicians during the first year of the COVID-19 pandemic. *J Healthc Manag*. 2024;69(5):368–386. DOI: <https://doi.org/10.1097/JHM-D-23-00124>
47. Harry EM, Carlasare LE, Sinsky CA, et al. Childcare stress, burnout, and intent to reduce hours or leave the job during the COVID-19 pandemic among US health care workers. *JAMA Netw Open*. 2022;5(7):e2221776. DOI: <https://doi.org/10.1001/jamanetworkopen.2022.21776>
48. Dyrbye LN, West CP, Sinsky CA, et al. Physicians' experiences with mistreatment and discrimination by patients, families, and visitors and association with burnout. *JAMA Netw Open*. 2022;5(5):e2213080. DOI: <https://doi.org/10.1001/jamanetworkopen.2022.13080>
49. Lyons NB, Bernardi K, Olavarria OA, et al. Gender disparity among american medicine and surgery physicians: A systematic review. *Am J Med Sci*. 2021;361(2):151–168. DOI: <https://doi.org/10.1016/j.amjms.2020.10.017>
50. Whaley CM, Koo T, Arora VM, Ganguli I, Gross N, Jena AB. Female physicians earn an estimated \$2 million less than male physicians over a simulated 40-year career. *Health Affairs*. 2021;40(12):1856–1864. DOI: <https://doi.org/10.1377/hlthaff.2021.00461>